

Factors, definitions, and citations for HECIA (Human Environment Connection & Interaction Atlas)

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Factors	Definitions	Citations
Acoustics	Acoustics refers to the noise profiles around the crew, usually perceptible noise. The acoustic environment includes continuous and intermittent noise. Continuous noise may be due to the environment or equipment, and typically measured with noise criteria (NC) curves that describe permissible levels (dB) of noise at given frequencies (hz) [Allen 2018]. Intermittent noise, defined as noise for 8 hours or less in a 24 hour period, has a ceiling of 115-dBA (weighted with time) as set by OSHA [US Occupational Safety & Health Administration 1983]. Noise may impact communication, nominal operations, and sleep [Allen 2018]. Acoustics can also include vibrations and other imperceptible noise.	Allen, C. S., Giraudo, M., Moratto, C., & Yamaguchi, N. (2018). Spaceflight environment. In <i>Space Safety and Human Performance</i> (pp. 87–138). Elsevier. Occupational Safety and Health Administration, US Department of Labor (1983). Occupational Noise Exposure: Hearing Conservation Amendment. Final Rule. 29 CFR 1910.95, Federal Register 46 (11), 4978–4981.
Action Processes	Action processes occur when teams are actively conducting activities that lead to goal achievement. Processes most salient during the action phase are as follows [Marks 2001]: - Monitor progress: tracking tasks and progress toward team goals. This includes providing feedback and updates to team members to evaluate progress and likelihood of success. - Systems monitoring: tracking status of resources and environmental conditions. This is also known as situation awareness, or monitoring of internal systems (e.g., equipment, resources) and external systems (e.g., weather) relevant to mission success. - Team monitoring: assisting team members to complete their tasks, including offering feedback, helping, or offloading the task. - Coordination: managing the sequence and timing of interdependent tasks. This information exchange and mutual adjustment of activities aligns the pace and sequencing of actions for more productive goal accomplishment.	Marks, M. A., John E. Mathieu, & Stephen J. Zaccaro. (2001). A Temporally Based Framework and Taxonomy of Team Processes. <i>The Academy of Management Review</i>, 26(3), 356–376.
Agreeableness	Tendency toward trust, straightforwardness, altruism, compliance, modesty, tender-mindedness [Matthews 2003].	Matthews, G., Deary, I. J., & Whiteman, M. C. (2003). <i>Personality Traits, SECOND EDITION</i>. Cambridge University Press.
Altered Gravity	Altered gravity refers to gravity (or acceleration) levels that are different than typical sea-level Earth gravity.	Norsk, P. (2021). Physiological effects of spaceflight–Weightlessness: An overview. <i>Handbook of Bioastronautics</i>, 93-105.

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	We denote the gravity level on Earth ($\sim 9.81 \text{ m/s}^2$ acceleration toward the Earth, depending on elevation) as 1 'g'. Altered gravity includes microgravity '0g' (e.g., freefall around Earth, as seen onboard the International space Station [DeLombard 2004]) and partial gravity: Martian '0.38g' (or about 1/3 Earth gravity) and Lunar '0.16g' (or about 1/6 Earth gravity) [Pletser 2012].	Pletser, V., Winter, J., Duclos, F., Bret-Dibat, T., Friedrich, U., Clervoy, J. F., ... & Sundblad, P. (2012). The first joint European partial-G parabolic flight campaign at Moon and Mars gravity levels for science and exploration. <i>Microgravity Science and Technology</i>, 24, 383-395.
Anxiety	Anxiety is characterized by tension, worried thoughts, and physiological responses (e.g., elevated blood pressure) [Corsini and Ozaki 1994]. We also wish to distinguish clinical anxiety disorder (e.g., Generalized Anxiety Disorder), subclinical anxiety is also relevant for behavioral health and performance [Judd et al. 2002]. Anxiety is both a state and trait, meaning that some people are more disposed to anxious feelings, while others may experience anxiety more acutely [Brown 2021]. Anxiety and depression also often co-occur [Kircanski et al. 2017]. Beyond clinical interviews or professional assessments, subjective measures of anxiety exists in the form of surveys; some widely-used ones are the State-Trait Anxiety Inventory (STAI), Beck Anxiety Inventory (BAI), and the Hospital Anxiety and Depression Scale - Anxiety (HADS-A) [Julian 2011]. The standard measure questionnaire for mood used by NASA, the Profile of Mood States (POMS), encompasses a tension-anxiety subconstruct [McNair 1971].	Brown, B. (2021). <i>Atlas of the heart: Mapping meaningful connection and the language of human experience</i>. Random House. Corsini, R. J., & Ozaki, B. D. (1994). <i>Encyclopedia of psychology</i> (Vol. 1). New York: Wiley. Julian, L. J. (2011). Measures of anxiety: state-trait anxiety inventory (STAI), Beck anxiety inventory (BAI), and Hospital anxiety and Depression scale-anxiety (HADS-A). <i>Arthritis care & research</i>, 63(S11). McNair, D. M., Lorr, M., & Droppleman, L. F. (1971). <i>Profiles of mood states</i>. San Diego: Educational and Industrial Testing Services. Judd, L. L., Schettler, P. J., & Akiskal, H. S. (2002). The prevalence, clinical relevance, and public health significance of subthreshold depressions. <i>Psychiatric Clinics</i>, 25(4), 685-698. Kircanski, K., LeMoult, J., Ordaz, S., & Gotlib, I. H. (2017). Investigating the nature of co-occurring depression and anxiety: Comparing diagnostic and dimensional research approaches. <i>Journal of affective disorders</i>, 216, 123-135.
Atmosphere	The atmosphere refers to the gases present available to the crew, usually in the environment, habitat, or specialized containers. The acceptable pressure range for indefinite human exposure is between 5-15 psia with at least 30% inert dilute gas, with the target normoxia range as 2.8-3.0 psia. Occasional short-term hypoxia or hyperoxia may be tolerated for EVA procedures. The operational limits for temperature and humidity is based on the ASHRAE 55 and anchored at 18C/75%RH at the upper left corner and 27C/25%RH at the lower right corner. The ideal ranges are 20-25C and 30-60%	Wieland, P.O. (2005). <i>Designing For Human Presence in Space: An Introduction to Environmental Control and Life Support Systems (ECLSS)</i>. NASA. NASA/TM—2005–214007. NOAA. (2024) Trends in Atmospheric Carbon Dioxide. National Oceanic and Atmospheric Administration Global Monitoring Laboratory. Accessed April 9, 2024.

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	relative humidity. Specific guidelines are available in NASA-STD-3001-Vol-1. Atmospheric considerations include pressure, gas composition, partial pressures of constituent gasses, absolute/relative humidity, temperature, microbes, trace contaminants, and particulate matter. In closed, pressurized, artificial atmospheres such as the one onboard the International Space Station, the atmosphere must be carefully monitored and managed [Wieland 2005]. Considerations for spacecraft includes flammability (limiting the concentration of oxygen), the choice of inert gas filler (some can alter vocal pitch), and total pressure to limit decompression operations to match spacesuit pressure (typically lower to allow range of motion). One of the main effects of climate change is the escalation of average global temperatures and increased CO2 concentration (currently over 425 ppm as of April 2024) [NOAA 2024]. These changes may create additional considerations for atmospheric management when designing for Earth.	NASA. (2023). NASA Spaceflight Human-System Standard Volume-1, Crew Health. NASA-STD-3001-Vol-1.
Autonomy	Autonomy is the ability to make independent decisions, often task-related (how and when to complete a task) in spaceflight settings. Closely related is autonomy, which refers to the capacity to act intentionally and/or the sense of volition (the power to enact one's will) [Schlosser 2015][Goemaere 2016]. Current Low Earth Orbit operational paradigms provide little autonomy to the crewmembers, as work schedules are set by mission control and modifications to procedures or plans are consulted with ground [Goemaere 2016]. However, current autonomy levels will need to change as mission distances and communication delays increase, leading to less ground support [Marquez 2017]. While there are efforts to increase crew autonomy through self-scheduling activities, for example, there are still major unknowns on the balance between crew and ground control [Marquez 2017].	Schlosser, M. (2015). Agency. Stanford Encyclopedia of Philosophy. Accessed May 2, 2024. Goemaere, S., Vansteenkiste, M., & Van Petegem, S. (2016). Gaining deeper insight into the psychological challenges of human spaceflight: The role of motivational dynamics. <i>Acta Astronautica</i>, 121, 130-143. Marquez, J. J., Hillenius, S., Kanefsky, B., Zheng, J., Deliz, I., & Reagan, M. (2017, March). Increasing crew autonomy for long duration exploration missions: self-scheduling. In 2017 IEEE Aerospace Conference (pp. 1-10). IEEE.
Boredom	Boredom is defined as the negative experience of wanting, but being unable to, engage in satisfying activity [Eastwood 2012]. Boredom arises from a lack of stimulation, attention, or arousal [Bench and Lench 2013]. The standard measure questionnaire for mood used by NASA, the Profile of Mood States (POMS),	Bench, S. W., & Lench, H. C. (2013). On the function of boredom. <i>Behavioral sciences</i>, 3(3), 459-472. Eastwood, J. D., Frischen, A., Fenske, M. J., & Smilek, D. (2012). The unengaged mind: Defining

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	encompasses a vigor-activity subconstruct, which may be a close approximate [McNair 1971].	boredom in terms of attention. Perspectives on psychological science, 7(5), 482-495. McNair, D. M., Lorr, M., & Droppleman, L. F. (1971). Profiles of mood states. San Diego: Educational and Industrial Testing Services.
Circulation	Circulation refers to the movement through space. Circulation spaces encompass movement during the “approach” and “entry” phases, before entering into the actual building or space [Ching 2023], which may be relevant for extravehicular missions (e.g., visible and compatible docking ports for vehicles). In the interior, considerations include “configurations of the path” (e.g., linear, radial, spiral) that dictates the sequence of spaces; and “path-space relationships”, which considers if paths run adjacent to spaces, cut through spaces, or terminate in a space [Ching 2023]. Circulation may be designed, and change over time. For instance, “desire paths” refer to dirt paths that form over time due to pedestrians bypassing sidewalks [Lidwell 2010].	Ching, F. D. (2023). Architecture: Form, space, and order. John Wiley & Sons. Lidwell, W., Holden, K., & Butler, J. (2010). Universal principles of design, revised and updated: 125 ways to enhance usability, influence perception, increase appeal, make better design decisions, and teach through design. Rockport Pub.
Cognitive Workload	Cognitive workload refers to the amount of cognitive and mental effort required to perform a task or a set of concurrent tasks [Hoedemaeker 2002]. Examples of cognitive workload measurement include subjective surveys (usually administered immediately after task completion), performance metrics (e.g., errors, time to task completion), and physiological measures (e.g., heart rate, pupil dilation) [Stanton 2005][Marshall 2002].	Hoedemaeker, M. (2002). Summary Description of Workload Indicators: WP1 Workload Measures. Human Machine Interface and the Safety of Traffic in Europe Growth Project. GRD1-2000-25361. HASTE. Institute for Transport Studies. Leeds, UK: University of Leeds. Stanton, N. (Ed.). (2005). Handbook of human factors and ergonomics methods. CRC Press. Marshall, S. P. (2002, September). The index of cognitive activity: Measuring cognitive workload. In Proceedings of the IEEE 7th conference on Human Factors and Power Plants (pp. 7-7). IEEE.
Communication Delay	The delay caused by the limit of speed of light and equipment latency. Missions rely on being able to communicate quickly with crew members in space. Communication delays for missions beyond Low Earth Orbit directly increase mission risks via increased time for resupply and emergency evacuations [Parisi 2023]. Seconds to minutes communication delay impacts real-time guidance from the ground. Lunar (5-14 seconds [Parisi 2023]) and Martian (6.5-44 min [Schuster 2007]) communication delays would require crew members	Parisi, M., Panontin, T., Wu, S.-C., Mctigue, K., & Vera, A. (2023). Effects of Communication Delay on Human Spaceflight Missions. 14th International Conference on Applied Human Factors and Ergonomics (AHFE 2023). Schuster, Alfons (2007): Intelligent Computing Everywhere. Springer, Berlin. P. 235-236.

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	to make independent decisions without relying on ground support [Parisi 2023].	
Confined	The degree of physical constraint into a specified volume or area. Confined environments, such as submarines, space stations, Antarctic stations, even our homes during COVID lockdowns, are joined by the commonality that there are physical and social constraints to leaving a certain space. Beyond limited mobility outside of the environment due to extreme environmental conditions, the indoor environment might also offer limited mobility [Nicolas 2016]. Confinement may also incorporate the access to resources within a walkable or accessible neighborhood. For example, the inequitable distribution of healthcare, grocery stores, libraries, and other social services; or the impossibility of gaining access to certain services in refugee camps, can contribute to the definition of confinement [Mouratidis 2020][Razum 2022].	Nicolas, M., Suedfeld, P., Weiss, K., & Gaudino, M. (2016). Affective, Social, and Cognitive Outcomes During a 1-Year Wintering in Concordia. <i>Environment and Behavior</i>, 48(8), 1073–1091. Mouratidis, K. (2020). Neighborhood characteristics, neighborhood satisfaction, and well-being: The links with neighborhood deprivation. <i>Land Use Policy</i>, 99, 104886. Razum, O., Dawson, A., Eckenwiler, L., Wild, V. (2022). Refugee Camps: Paradise or Purgatory?. In: Razum, O., Dawson, A., Eckenwiler, L., Wild, V. (eds) <i>Refugee Camps in Europe and Australia</i>. Palgrave Macmillan, Cham.
Conscientiousness	Tendency toward competence, order, dutifulness, achievement striving, self-discipline, deliberation [Matthews 2003].	Matthews, G., Deary, I. J., & Whiteman, M. C. (2003). <i>Personality Traits, SECOND EDITION</i> . Cambridge University Press.
Crew Performance	Individual-level performance is typically task-related (e.g., accomplishment of task goals, errors, task duration). However, most often tasks are accomplished as a team [Bell 2015], and overall performance relies on both the individual team members and the team as a whole [Mathieu 2014]. Team performance comprises the health of a team and the individual performances of each member. Based on the input-mediator-output-input model [Ilgen 2005][Grossman 2017], team performance here encompasses both the accomplishment of tasks and goals as set by mission parameters, and also the cognitive and affective “emergent” states of a team [Grossman 2017][Marks 2001]. Affective states include team cohesion, team confidence, and team trust; cognitive states include team mental models, situational awareness, and team learning [Grossman 2017]. In isolated environments, the effect of team cohesion has an exaggerated relationship with team performance due to the extreme environment [Vessey and Landon 2017].	Bell, S. T. et al. (2015). Critical Team Composition Issues for Long-Distance and Long-Duration Space Exploration. Mathieu, J. E., Tannenbaum, S. I., Donsbach, J. S., & Alliger, G. M. (2014). A Review and Integration of Team Composition Models: Moving Toward a Dynamic and Temporal Framework. <i>Journal of Management</i>, 40(1), 130-160. Ilgen, D. R. et al. (2005). Teams in organizations: From input–process–output models to IMOI models. <i>Annual Review of Psychology</i>, 56, 517–543. Grossman, R. et al. (2017). Teamwork Processes and Emergent States. In E. Salas et al., <i>The Wiley Blackwell Handbook of the Psychology of Team Working and Collaborative Processes</i> (p. 243–269). Wiley. Marks, M.A. et al. (2001). A Temporally Based Framework and Taxonomy of Team Processes. <i>The Academy of Management Review</i>, 26(3), 356–376.

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		Vessey, W. B., & Landon, L. B. (2017). Team Performance in Extreme Environments. In E. Salas, R. Rico, & J. Passmore (Eds.), <i>The Wiley Blackwell Handbook of the Psychology of Team Working and Collaborative Processes</i> (1st ed., pp. 531–553). Wiley.
Crew Size	The number of people in a crew. In US space missions, programs have differed in the crew size but consistently remain a small group of crewmembers. From Mercury (1) to Gemini (2), Apollo (3), Skylab (3), Space Shuttle (6-8), and ISS (7), it's most likely that in the foreseeable future the crew size will remain in this range [Vakoch 2013][NASA 2024]. These figures do not typically include those who are supporting the mission, including ground support, family, and friends. When considering an Earth analogy to space missions, we can consider immigrants and refugees, who are similarly displaced from home (either voluntarily or involuntarily). In Earth-based contexts, the United Nations currently (2020) estimate there are 281 million immigrants world-wide, which is about 3.6% of the global population [Mcauliffe and Triandafyllidou 2022]. The global refugee population exceeds 30 million individuals, and more than 100 million are forcibly displaced (including refugees) due to persecution, conflict, violence, and human rights violations [Masterson 2023][UNHCR 2024]. Sizes of immigrant groups may vary from individual, to family, to 1000 refugees arriving in a single city [Pottie-Sherman and Wilkes 2017].	Vakoch, D. A. (Ed.). (2013). <i>On Orbit and Beyond: Psychological Perspectives on Human Spaceflight</i>. Springer Berlin Heidelberg. NASA. (2024) “Station Overview” https://www.nasa.gov/international-space-station/space-station-overview/. Accessed March 12, 2024. Mcauliffe, Marie and Anna Triandafyllidou. (2021). <i>World migration report 2022</i>. International Organization for Migration (IOM). Masterson, D. (2023). Refugee Networks, Cooperation, and Resource Access. <i>American Political Science Review</i>, 1–17. UNHCR. (2024) “About UNHCR” https://www.unhcr.org/about-unhcr Accessed March 12, 2024. Pottie-Sherman, Y. and Wilkes, R. (2017), Does Size Really Matter? On the Relationship between Immigrant Group Size and Anti-Immigrant Prejudice†. <i>Int Migr Rev</i>, 51: 218-250.
Curiosity	Curiosity refers to the desire for more knowledge. Curiosity can be a state (i.e., a curious person) or a trait (i.e., feel curious in the moment), and has both cognitive and affective components [Brown 2021]. Interest, or the intellectual component of curiosity, is a prerequisite to the eventual recognition of a knowledge gap and investment to fill the gap through exploration and learning [Brown 2021].	Brown, B. (2021). <i>Atlas of the heart: Mapping meaningful connection and the language of human experience</i>. Random House.
Density	Density is related to the amount of people in a given space.	Fleming, I., Baum, A., & Weiss, L. (1987). Social density and perceived control as mediators of crowding stress in high-density residential

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	Density can be increased by adding more people into a given space (increasing social density) or decreasing the size of the space (increasing spatial density) [Fleming 1987].	neighborhoods. <i>Journal of Personality and Social Psychology</i> , 52(5), 899–906.
Depression	Depression is a cluster of behaviors that are persistent in time, including lack of interest in activities, loss of appetite, negative self-view, and alteration of mood (e.g., sadness, loneliness, apathy) [Brown 2021][Beck and Alford 2009]. Subclinical symptoms (e.g., mild and/or transient sadness), are prevalent in non-clinical populations (a n = 9160 study found a 1-year prevalence of 8.6%) and also relevant for behavioral health [Judd et al. 2002]. Common scales for measuring depression include the Beck Depression Inventory (BDI) and the Depression Anxiety Stress Scales (DASS) [Beck 1987][Lovibond and Lovibond 1995]. The standard measure questionnaire for mood used by NASA, the Profile of Mood States (POMS), encompasses a depression-dejection subconstruct [McNair 1971]. While sadness is common to depression, it is not a necessary trait of depression [Beck and Alford 2009]. Although we have not defined it explicitly, it is an important component of the human emotional spectrum, with its most salient representation in behavioral health as a symptom of depression. Similarly, grief - composed of feelings of loss and longing – may be associated with ‘depressed mood’ but is realistically distinct [Brown 2021].	Beck, A. T., & Alford, B. A. (2009). <i>Depression: Causes and treatment</i>. University of Pennsylvania Press. Brown, B. (2021). <i>Atlas of the heart: Mapping meaningful connection and the language of human experience</i>. Random House. Beck, A. T., Steer, R. A., & Brown, G. K. (1987). <i>Beck depression inventory</i>. San Antonio, TX. Lovibond, P. F., & Lovibond, S. H. (1995). The structure of negative emotional states: Comparison of the Depression Anxiety Stress Scales (DASS) with the Beck Depression and Anxiety Inventories. <i>Behaviour research and therapy</i>, 33(3), 335-343. McNair, D. M., Lorr, M., & Droppleman, L. F. (1971). <i>Profiles of mood states</i>. San Diego: Educational and Industrial Testing Services. Judd, L. L., Schettler, P. J., & Akiskal, H. S. (2002). The prevalence, clinical relevance, and public health significance of subthreshold depressions. <i>Psychiatric Clinics</i>, 25(4), 685-698.
Distance From Home/Earth	In general applications, distance from home is dependent on the changing definition of home, which has been hypothesized to generally fall into categories of physical structures, territories, self and self-identity, and social-cultural units [Hayward 1975][Case 1996]. Therefore, “home” may be multiple, constructed, and changing as one leaves and creates new homes. Distance from home can be considered as difficulties due to physical constraints (distance, time), economic (unemployment, cost), social-cultural (values), or political (visas, policies, asylum, deportation) [Angulo 2010]. Space-specific considerations include orbit alignment, mission impacts, and resource limitations (food, water, fuel).	Hayward, G. (1975). Home as an environmental and psychological concept. <i>Landscape</i>, 20, 2–9. Case, D. (1996). CONTRIBUTIONS OF JOURNEYS AWAY TO THE DEFINITION OF HOME: AN EMPIRICAL STUDY OF A DIALECTICAL PROCESS. <i>Journal of Environmental Psychology</i>, 16(1), 1–15. https://doi.org/10.1006/jevp.1996.0001. Angulo, M. I. (2010). WHEN HOME IS NO LONGER THERE: RETURN MIGRATION IN A TIME OF CRISIS. <i>Migration</i>, 48.
Environmental Control	The monitoring and maintenance of environmental parameters needed to create a livable environment.	Broyan, J. L., Shaw, L., McKinley, M., Meyer, C., Ewert, M. K., Schneider, W. F., Meyer, M., Ruff, G. A., Owens, A. C., & Gatens, R. L. (2021). <i>NASA Environmental Control and Life Support</i>

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	Requirements for a liveable atmosphere are described under 'Atmosphere'. Further guidelines for toxic trace contaminants and potable water are described in NASA-STD-3001-Vol-1. Recommended drinking water intake is described under 'Food'. Recommended potable water quantity for personal hygiene is 400 mL/day (quantities for medical events and reentry events are in the standards). Environmental monitoring and control is necessary to maintain a habitable environment inside a confined space. For a spacecraft, the subsystem responsible for regulating the environment is termed “Environmental Control and Life Support Systems”, or ECLSS. ECLSS comprises many subcomponents, which are responsible for thermal management, maintaining and monitoring atmospheric content (pressure, partial pressures, moisture, microbes, contaminants), water, and waste. It also typically includes fire detection, suppression, and cleanup systems [Broyan 2021].	Technology Development for Exploration: 2020 to 2021 Overview. 50th International Conference on Environmental Systems. NASA. (2023). NASA Spaceflight Human-System Standard Volume-1, Crew Health. NASA-STD-3001-Vol-1.
Environmental Monotony	Environmental monotony is associated with unchanging characteristics of the built and natural environment, including layout, lighting, colors, textures, views, and sensory inputs. In general, monotony is described by an unchanging environment, repetition of events or situations, and chronic absence of novelty [Das 2015].	Das, S.K., Barhwal, K., Hota, S.K. (2015). Disrupting monotony during social isolation stress prevents early development of anxiety and depression like traits in male rats. BMC Neurosci 16, 2.
Ergonomics	Ergonomics integrates knowledge about the human physical and cognitive limitations with the design of interfaces, products, environments, and jobs [Nussbaum and Van Dieën 2005]. For spaceflight, general design standards that consider strength, muscle effectiveness and fatigue, aerobic capacity, and anthropometry can be found in NASA-STD-3001-Vol-1 (Human-System Standard), NASA/TP-2014-218556 (Human Integration Design Process), OCHMO-HB004 (Anthropometry, Biomechanics, and Strength). Earth-based standards are numerous, including workplace safety guidelines, building and architecture codes (varies by location), and more specific design guides. Physical ergonomics focuses on the demands placed on the human and the capabilities of the human to fulfill these, to ensure the	Nussbaum, M. A., & Van Dieën, J. H. (2005). Physical Ergonomics. In M. Kutz (Ed.), Mechanical Engineers' Handbook (1st ed., pp. 762–781). Wiley. Stanton, N. (Ed.). (2005). Handbook of human factors and ergonomics methods. CRC Press. Churchill, E., Laubach, L. L., Mcconville, J. T., & Tebbetts, I. (1978). Anthropometric Source Book—Volume I; Anthropometry for Designers.pdf. NASA. 19790003563 Hollnagel, E. (1997). Cognitive ergonomics: It's all in the mind. Ergonomics, 40(10), 1170–1182.

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	demands do not exceed the capabilities [Nussbaum and Van Dieën 2005]. One example is anthropometry, which is a science that characterizes the human body and a design tool for physical ergonomics; NASA has published an Anthropometric Source Book based largely on data from the military population [Churchill 1978]. Cognitive ergonomics focuses on the cognitive resources and demands for a given task, environment, and user [Stanton 2005]. The quality of work is affected by the user's understanding of the situation (e.g., situation awareness, mental models, goals), which in turn affects the quality of work [Hollnagel 1997]. One key area of application is in human-robot interaction and joint cognitive systems, where cognitive ergonomics provides a lens of the "fit" of the work and mind in a coupled system, as opposed to simply the effects of one on the other [Hollnagel 1997].	
Exercise Equipment	Exercise equipment are tools and machines whose primary function is to provide an exercise environment or program. Early space missions used elastic bands coupled with physical exercises as the primary countermeasure against musculoskeletal deconditioning [Scott 2023]. Currently, the International Space Station exercise equipment includes the Advanced Resistive Exercise Device (ARED), a treadmill (T2), and a cycle ergometer (CEVIS) [Scott 2023].	Scott, J. M., Feiveson, A. H., English, K. L., Spector, E. R., Sibonga, J. D., Dillon, E. L., Ploutz-Snyder, L., & Everett, M. E. (2023). Effects of exercise countermeasures on multisystem function in long duration spaceflight astronauts. <i>Npj Microgravity</i> , 9(1), 11.
Extraversion	Tendency toward warmth, gregariousness, assertiveness, activity, excitement seeking, positive emotions [Matthews 2003].	Matthews, G., Deary, I. J., & Whiteman, M. C. (2003). <i>Personality Traits, SECOND EDITION</i> . Cambridge University Press.
Extreme	The degree of hostility of the external physical environment. Hostile environments exist on Earth and in space. The Antarctic region, characterized by extreme cold (often dipping below -100F/-73C), extremely variable daylight hours with 5-6 months of complete darkness, low humidity, high altitude, and hazards (blizzards, crevices, ice), is often used as an environmental analog for outer space [Sandal 2018][Harrison 1991][Palinkas and Suedfeld 2008][Salam 2012]. The space environment, further characterized by vacuum, microgravity, temperature swings (due to fluctuations between receiving solar heat and emptying heat to the coldness of space), micrometeoroids, space debris, altered light/dark cycles, and radiation [Thirsk 2009].	Sandal, G. M., van deVijver, F. J. R., & Smith, N. (2018). Psychological Hibernation in Antarctica. <i>Frontiers in Psychology</i> , 9, 2235. Thirsk, R., Kuipers, A., Mukai, C., & Williams, D. (2009). The space-flight environment: The International Space Station and beyond. <i>Canadian Medical Association Journal</i> , 180(12), 1216–1220. Harrison, A. A., Clearwater, Y. A., & McKay, C. P. (Eds.). (1991). <i>From Antarctica to Outer Space: Life in Isolation and Confinement</i> . Springer New York.

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		Palinkas, L. A., & Suedfeld, P. (2008). Psychological effects of polar expeditions. <i>The Lancet</i>, 371(9607), 153–163. Salam, A. P. (2012). Exploration class missions on earth: Lessons learnt from life in extreme Antarctic isolation and confinement. In A. Chouker (Ed.), <i>Stress challenges and immunity in space</i> (pp. 425-439). Berlin, Germany: Springer.
Family and Friends	Family and friends are close interpersonal relationships that provide social support.	
Fatigue	Fatigue is multidimensional, encompassing physical, physiological, and psychological aspects. Fatigue is defined as “the awareness of a decreased capacity for physical and/or mental activity due to an imbalance in the availability, utilization, and/or restoration of resources needed to perform activity” [Aaronson 1999]. Fatigue is one of the most often reported symptoms by astronauts [Barger 2014]. The standard measure questionnaire for mood used by NASA, the Profile of Mood States (POMS), encompasses a fatigue-inertia subconstruct [McNair 1971]. A subclinical manifestation of fatigue relevant to operational contexts is 'neurasthenia', defined as 'a nervous or mental weakness manifesting itself in tiredness... and quick loss of strength, low sensation threshold, extremely unstable moods, and sleep disturbance' [Kanas and Manzey 2008]. Reports of hostility or irritability without meeting clinical thresholds of anxiety or depression has been reported in spaceflight [Dev 2023].	McNair, D. M., Lorr, M., & Droppleman, L. F. (1971). <i>Profiles of mood states</i>. San Diego: Educational and Industrial Testing Services. Aaronson, L. S., Teel, C. S., Cassmeyer, V., Neuberger, G. B., Pallikkathayil, L., Pierce, J., ... & Wingate, A. (1999). Defining and measuring fatigue. <i>Image: the journal of nursing scholarship</i>, 31(1), 45-50. Barger, L. K., et al. (2014). Prevalence of sleep deficiency and use of hypnotic drugs in astronauts before, during, and after spaceflight: an observational study. <i>The lancet neurology</i>, 13(9), 904–912. Kanas, N., & Manzey, D. (2008). <i>Space psychology and psychiatry</i> (2nd ed). Microcosm Press ; Springer. Dev, S. I., Dotson, V., Mulavara, A., Nelson, G. A., Schorn, J. M., Whitmire, A. M., Whiting, S. E., & Williams, T. J. (2023). <i>Evidence Report: BMed. NASA</i>.
Fixtures	Here, we define fixtures to include assets of the habitat that are installed, fixed, and/or (semi-)permanent. Some examples include apertures (i.e., openings such as doors, windows, docking ports), signage (e.g., node labels on the International Space Station), stowage (e.g., spaces for personal items and photos), and furniture. Fixtures can serve function and decorative purposes for the habitat. Note that while exercise equipment may fall	

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	under “fixtures”, but we have defined it separately due to its significance in the space paradigm.	
Food	Food is consumables containing the six types of nutrients: lipids, vitamins, water, carbohydrates, protein, and minerals [Insel 2014]. The recommended caloric intake for females (males) is around 2,000 (2,700) kcal per day; and 2.7 (3.7) liters of water per day; NASA provides a minimum of 2.5L/day [NASA 2023]. The recommendation fluctuates on activity levels and physical condition. A balanced diet includes carbohydrates (50% of daily caloric intake), proteins (20% of daily caloric intake; or 1.6g/kg body weight), fats (25% of daily caloric intake; with saturated fats <10%), fiber (25 (28) g/day), and a variety of minerals and vitamins [Faizan and Rouster 2023]. Food is the primary source of nutrition. Food choices may be due to personal preferences, availability, sensory influences (taste, texture, smell), and economics [Insel 2014]. Food supplies essential nutrients, providing energy, hydration, vitamins, minerals, carbohydrates, and proteins [Insel 2014]. Considerations for the space food system include safety, nutrition, usability, resource minimization, reliability, stability, variety, and palatability [Douglas 2020]. These sets of considerations may be similar for environments burdened by resource limitations, such as for disaster-relief.	Insel, P. M. (2014). Nutrition. Jones & Bartlett Publishers. Douglas, G. L., Zwart, S. R., & Smith, S. M. (2020). Space food for thought: Challenges and considerations for food and nutrition on exploration missions. <i>The Journal of Nutrition</i>, 150(9), 2242-2244. Faizan, U., & Rouster, A. S. (2023). Nutrition and hydration requirements in children and adults. StatPearls Publishing. NASA. (2023). NASA Spaceflight Human-System Standard Volume-1, Crew Health. NASA-STD-3001-Vol-1.
Genetics	Genetics is a branch of biology dealing with heredity and variation. Here, we focus on the chromosomal theory of genetics, which is that genetic differences and similarities are inherited through DNA replication and therefore inherits genetic continuity through generations [Krug 2006]. Phenotypic (i.e., observable) differences may account for traits such as eye color or handedness [Krug 2006]. However, while more complex traits such as predisposition to illness could be partially explained by phenotypic differences, they may also differ by environmental influences, age, and interaction effects between genetics and these variables [Kujala 2011]. Genetic differences and their predictability and effects on personality are still active areas of research, and multiple theories exist on genetic, environmental, and interactive (genetic and environmental) explanations for personality [Matthews 2003] – which has implications for behaviors, habits, and health.	Bouchard, T. J. (1994). Genes, Environment, and Personality. <i>Science</i>, 264(5166), 1700–1701. Matthews, G., Deary, I. J., & Whiteman, M. C. (2003). <i>Personality Traits, SECOND EDITION</i>. Cambridge University Press.

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Ground Support	Support that comes from an external resource (usually a consistent community of trained staff) for the crew. Spaceflight is an incredibly demanding effort and supported by an enormous amount of facilities, technology, and people on the ground. Each of the major space agencies with astronauts on the International Space Station has mission control and research/technology centers, comprising highly trained experts that support the challenges each autonomy may face [NASA 2023].	NASA. “Ground Facilities”. Updated Dec 28, 2023. Accessed April 30, 2024. https://www.nasa.gov/international-space-station/ground-facilities/
Group Living	The skills and context associated with living with others. In the context of spaceflight and other extreme environments, group living skills or competency refer to a unique subset of skills that apply to colleagues who work and live together [Landon 2024]. They become more salient as duration, isolation, and confinement increase [Landon 2024]. Group living skills for astronauts are essentially teaming skills that extend beyond work, into life situations. For example, ability to engage in interpersonal conversations constructively and not defensively; ability to identify and manage differences in personality, knowledge, cultural expectations; and ability to cooperate, take accountability, and assign responsibility [Landon and Paoletti 2020].	Landon, L. B., Miller, J. C., Bell, S. T., & Roma, P. G. (2024). When people start getting real: The Group Living Skills Survey for extreme work environments. <i>Frontiers in Psychology</i>, 15, 1348119. Landon, L. B., & Paoletti, J. (2020). Extreme roommates: Exploring group-living skills as a unique team skill area. In <i>Psychology and human performance in space programs</i> (pp. 217-235). CRC Press.
Habitable Volume	The volume available and accessible to the crew. When considering a pressurized system, there are differences between the total pressurized volume (the interior volume determined by the inner physical envelope) and the usable volume by its occupants [Rudisill 2010]. After the equipment, stowage, and outfitting of the interior, the remaining space is typically termed the “sand” volume (i.e., volume if sand could fill all crevices) [Rudisill 2010]. However, not all sand volume is usable or relevant for day-to-day living; for example, consider the volume beneath a couch or inside the fridge. Therefore, a smaller “net habitable volume” is typically the figure of merit for habitat design, and traditionally about 60% of total pressurized volume for space habitats [Rudisill 2010].	Rudisill, M., Howard, R., Griffin, B., Green, J., Toups, L., Kennedy, K. (2008). 'Lunar Architecture Team – Phase 2 Habitat Volume Estimation: ‘Caution When Using Analogs’', ASCE Earth and Space Conference, Long Beach, CA.
Individual Traits	Trait factors that are specific and (to an extent) invariant to a person. Individual traits accounting for variation, including genetics and personality. Personality traits are based on Costa and McCrea’s five-	

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	factor model of personality. Although no consensus for personality models exists, Costa and McCrea's Big Five model is heavily based on empirical research and widely used and adapted as a measurement scale (NEO-PI-R, IPIP-NEO, IPIP-NEO-120 [used by NASA]).	
Interpersonal Processes	Interpersonal processes occur through both transition and action phases, and lay the foundation for the effectiveness of other team processes [Marks 2001]. They are as follows [Marks 2001]: - Conflict management: helps resolve or manage conflict in two ways: 1) preemptively, by establishing conditions to prevent, control, or guide conflicts (e.g., establishing team norms), and 2) reactively, by working through task or interpersonal conflicts that have arisen (e.g., problem-solving, compromising). - Motivating: preserving a collective sense of confidence, motivation, and cohesion toward the team goal. - Affect management: calibrating and regulating team members' emotions that may be shifting due to task-based (e.g., stress), personal (e.g., animosity), and external (e.g., job security) factors.	Marks, M. A., John E. Mathieu, & Stephen J. Zaccaro. (2001). A Temporally Based Framework and Taxonomy of Team Processes. <i>The Academy of Management Review</i> , 26(3), 356–376.
Isolated	The degree of social isolation and separation away from sources of support or others. In isolated environments, physical/social interaction and telecommunication with usual sources of support (family, friends) are seriously disrupted [Nicolas 2016]. In times of crisis, evacuation from an isolated environment may be difficult, if not impossible [Sandal 2006].	Nicolas, M., Suedfeld, P., Weiss, K., & Gaudino, M. (2016). Affective, Social, and Cognitive Outcomes During a 1-Year Wintering in Concordia. <i>Environment and Behavior</i>, 48(8), 1073–1091. Sandal, G. M., Leon, G. R., & Palinkas, L. (2006). Human challenges in polar and space environments. <i>Reviews in Environmental Science and Bio/Technology</i>, 5(2–3), 281–296.
Kinship	Kinship, whose definition has been debated in anthropology for over 150 years [Sahlins 2011], is chosen here to be the encompassing umbrella of mutual relational closeness, which comprises several subconstructs: belonging, connection, and loneliness. During its conception, kinship was used to describe familial or genealogical ties, for example in “next of kin”. David Durkheim, the founder of sociology as a formal discipline, sought to explain kinship as “mutual relations of being, participation in one another's existence” [Sahlins 2011]. Belonging – the connection that arises between people and groups without the need to compromise	Sahlins, M. (2011). What kinship is (part one). <i>Journal of the Royal Anthropological Institute</i>, 17(1), 2-19. Brown, B. (2021). <i>Atlas of the heart: Mapping meaningful connection and the language of human experience</i>. Random House. Maslow, A., & Lewis, K. J. (1987). Maslow's hierarchy of needs. <i>Salenger Incorporated</i>, 14(17), 987-990.

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	authenticity or values [Brown 2021] – is a core concept to kinship. Meaningful connections, that allows people to feel seen and valued, are psychologically critical [Maslow and Lewis 1987]. Loneliness is the absence of such meaningful connections, and is associated with a greater risk of physiological and psychological illnesses [Brown 2021].	
Layout	Layout refers to the arrangement of spaces and the relationships (e.g., spatial, acoustic) between them, also sometimes referred to as 'adjacencies'. Typically, layouts for space habitats are arranged programmatically (i.e., by “programs” or primary activities supported by the space). For instance, one study organized activities that had taken place in space habitats (from Apollo to the International Space Station) into the following programs: sleep, hygiene, food, work, and leisure. Subsystems can also have specific layouts. For example, the International Space Station uses “racks” as a standard unit so systems (e.g., CO2 removal scrubber) and experiments can share standardized interfaces [Pelfrey and Jordan 2008].	Pelfrey, J., & Jordan, L. (2008). An EXPRESS Rack overview and support for microgravity research on the International Space Station (ISS). In 46th AIAA Aerospace Sciences Meeting and Exhibit (p. 819).
Life/Mission Events	Events that are not associated with planned everyday tasks (e.g., science, hygiene, meals); also contingency, emergency, and unplanned scenarios. These may include launch/landing, extravehicular activity, crew docking and transfer, injury, and conflicts. We distinguish demands as expected, chronic, repetitive operations; as opposed to events, which are more acute and may require more planning or immediate attention to resolve. The distinction between demands and events vary by context; for instance, in an hospital emergency department, what may be considered 'emergency events' are likely part of the everyday demand.	
Light/Dark Cycles	Light/dark cycles refer to the proportion and timing of light and darkness. Natural light/dark cycles typically occur every 24 hours on Earth with shifting proportions of light and dark, depending on where one resides and the time of year. Polar regions on Earth experience 6 months of day and 6 months of darkness, also known as polar day and night [Palinkas and Suedfeld 2008]. On the International Space	Palinkas, L. A., & Suedfeld, P. (2008). Psychological effects of polar expeditions. <i>The Lancet</i>, 371(9607), 153–163. Thirsk, R., Kuipers, A., Mukai, C., & Williams, D. (2009). The space-flight environment: the International Space Station and beyond. <i>Cmaj</i>, 180(12), 1216-1220.

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	Station, light/dark transitions occur every 45 minutes due to the velocity of the station in its orbit, resulting in 16 sunrises and sunsets every day [Thirsk 2009].	
Lighting	Lighting refers to the amount, placement, and type of illumination provided. NASA Standard NASA-STD-3001-Vol-1 describes minimum illumination requirements (horizontal ambient ≥ 350 lux; vertical ambient ≥ 200 lux; medical/maintenance ≥ 750 lux) between 2700-6500K. For circadian entrainment, general ambient lighting should be at a minimum of 250 melanoptic EDI lux, with sleep around 8 meEDI lux. While natural lighting is important for everyday life, it is limited in extreme environments. Here, we discuss artificial lighting created to support living and working tasks. Some considerations for lighting include color (wavelength on the visible spectrum), color temperature (in degrees Kelvin), and brightness (lumens) [Karlen 2017]. Operationally, additional design features such as dimmability, adjustments throughout the day to support circadian rhythms, and design of lighting layers (daylight, decorative, focal, task, and ambient) can be considered [Karlen 2017].	Karlen, M., Spangler, C., & Benya, J. R. (2017). Lighting design basics. John Wiley & Sons. NASA. (2023). NASA Spaceflight Human-System Standard Volume-1, Crew Health. NASA-STD-3001-Vol-1.
Media Attention	The amount of public and media (e.g., news outlets, social media) associated with an event, mission, or activity. Media attention is particularly important to the spaceflight population and other populations who may be in isolated, confined, and extreme environments (e.g., refugees, civilians in war zones). Since 1958, when NASA was chartered with providing the “widest practicable and appropriate dissemination of information concerning its activities”, NASA has maintained a public presence in the media [Makemson 2009]. Since 2009 with Mike Massimino's first ‘tweet’ from space, it has become common practice for astronauts to have public social media accounts. In most space-faring countries, astronauts are popular figures (as a group and some in their own right) with important public relations duties [Hersch 2012].	Makemson, H. (2009). Media, NASA, and America's Quest for the Moon (Vol. 4). Peter Lang. Hersch, M. H. (2012). Public Space. In <i>Inventing the American Astronaut</i> (pp. 131-158). New York: Palgrave Macmillan US.
Medical Capability	Medical capability is the “complete set of resources that enables the crew to perform medical monitoring, diagnosis, and treatment of medical conditions” [Antonsen 2022].	Antonsen, E. L., Myers, J. G., Boley, L., Arellano, J., Kerstman, E., Kadwa, B., Buckland, D. M., &

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	Estimating the appropriate range of medical supplies, resupplies, and equipment is an open topic as we have incomplete information on deconditioning and medical needs for deep-space missions. Probabilistic models, such as the NASA Integrated Medical Model, provides a guideline for medical resources [Antonsen 2022]. These may include tools, medicine, equipment, (crew and ground support) knowledge, training, references, and other resources. Medical capabilities should adequately match medical needs. However, prediction of medical capability requirements for untested mission profiles (e.g., Lunar or Martian mission) are difficult; medical needs are attributed by a complex web of factors, may be acute or chronic, and could develop/escalate as comorbidities [Antonsen 2022].	Van Baalen, M. (2022). Estimating medical risk in human spaceflight. <i>Npj Microgravity</i>, 8(1), 8.
Mission Demands	Planned operations and activities. Mission demands vary based on the specific mission profile and goals. This includes cognitive and physical demands that are part of the mission profile, including science, maintenance, hygiene, exercise, and meals. Demands also include performance pressure from ground control or operations. We distinguish demands as expected, chronic, repetitive operations; as opposed to events, which are more acute and may require more planning or immediate attention to resolve. The distinction between demands and events vary by context; for instance, in an hospital emergency department, what may be considered 'emergency events' are likely part of the everyday demand.	
Mission Duration	The total time length of a mission. Space mission durations vary, with the longest space mission being on the order of 1 year (Valeri Polyakov on Mir - 437 days; Frank Rubio on ISS - 371 days; Mark Vande Hei on ISS - 355 days) [Archie 2023]. Analog missions aim to replicate space-like conditions, such as isolation for 45 days in the Human Research Exploration Analog [NASA 2024]. Mission durations in the Antarctic region vary between the summer cohorts (about 3 months) and the winter-over cohorts (ranging from 6-9 months) [Harrison 1991].	Archie, A. (2023). "A NASA Astronaut is back on Earth after a year in space, the longest for an American." Updated September 28, 2023. Accessed March 12, 2024. NASA. (2024) "HERA" Accessed March 12, 2024. https://www.nasa.gov/mission/hera/. Harrison, A. A., Clearwater, Y. A., & McKay, C. P. (Eds.). (1991). <i>From Antarctica to Outer Space: Life in Isolation and Confinement</i>. Springer New York.

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	The duration of voluntary migrations can be up to multigenerational. The duration of involuntary migrations can vary, from weeks (Kosovars in 1999) to decades (Palestinians, 70 years) [Devictor 2017].	
Neuroticism	Tendency toward anxiety, angry hostility, depression, self-consciousness, impulsiveness, vulnerability [Matthews 2003].	Matthews, G., Deary, I. J., & Whiteman, M. C. (2003). <i>Personality Traits, SECOND EDITION</i> . Cambridge University Press.
Nostalgia/Homesickness	Homesickness is the set of feelings that are associated with the loss of known people and places [Brown and Perkins 1992]. Nostalgia is difficult to define precisely, but one interpretation is a primarily positive, context-specific emotion that combines sadness, happiness, yearning, and loss [Brown 2021]. We grouped them here since they are likely to occur in parallel in the context of isolated, confined, and extreme environments.	Brown, B. B., & Perkins, D. D. (1992). Disruptions in Place Attachment. In I. Altman & S. M. Low (Eds.), <i>Place Attachment</i> (pp. 279–304). Springer US. Brown, B. (2021). <i>Atlas of the heart: Mapping meaningful connection and the language of human experience</i> . Random House.
Openness	Tendency toward fantasy, aesthetics, feelings, actions, ideas, values [Matthews 2003].	Matthews, G., Deary, I. J., & Whiteman, M. C. (2003). <i>Personality Traits, SECOND EDITION</i> . Cambridge University Press.
Physical Health	Health is a large umbrella term. A typical consideration is the length of life, expressed in average life expectancy, mortality rates, and death due to specific causes [NCHS 2024]. Physical health metrics may include physical ability, physical limitations, pain, and health outlook [Ware 1987]. Spaceflight considerations include adverse effects to radiation (e.g., increased cancer risk, degenerative tissue diseases), central nervous system deconditioning (including sensorimotor and neurovestibular impacts), muscle/bone/cardiovascular atrophy due to unloading/migravity, inadequate food and nutrition, lunar dust exposure, and injuries due to extravehicular activities [McPhee and Charles 2009]. On Earth, the United States Center for Disease Control recommends 30 minutes of moderate exercise every day to maintain health [Pate 1995].	NCHS: Health. (2024). United States: U.S. Department of Health and Human Services. Accessed May 5, 2024. Ware, E. J. (1987). Standards for validating health measures: Definition and content. <i>Journal of Chronic Diseases</i> , 40(6), 473–480. McPhee, J. C., & Charles, J. B. (Eds.). (2009). <i>Human health and performance risks of space exploration missions: evidence reviewed by the NASA human research program</i> (Vol. 3405). US National Aeronautics & Space Administration. Pate, R. R., Pratt, M., Blair, S. N., et al. (1995). Physical activity and public health: a recommendation from the Centers for Disease Control and Prevention and the American College of Sports Medicine. <i>Jama</i> , 273(5), 402-407.
Physical Workload	Physical workload refers to the amount, duration, and type physical work being performed. Physical activity is necessary for sustaining health. The American Heart Association recommends at least 150 minutes of moderate (50-	Stanton, N. (Ed.). (2005). <i>Handbook of human factors and ergonomics methods</i> . CRC Press. NASA. (2025). <i>Astronaut physiological deconditioning and exercise prescription</i>

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	70% of max heart rate) activity or 75 minutes of vigorous activity (70-85% of max heart rate) per week. Current spaceflight exercise protocol of 2.5hr/day, 6 days/week includes aerobic exercise (around 30 min at 70-100% VO2max) and resistance training (30-60 min of a pre-determined % of body weight), and progresses in intensity as appropriate [NASA 2025]. It can be measured in many ways, including physical assessments (e.g., musculoskeletal discomfort surveys, muscle fatigue assessments), physiological measurements (e.g., heart rate, respiration, electromyography), and behavioral measurements (e.g., observation) [Stanton 2005].	countermeasures in spaceflight. NASA/SP-20250000273.
Place Attachment	Place attachment is the affective bond and the process of bonding (i.e., “attachment”) between an individual and a space of significance (i.e., “place”) [Altman and Low 1992][Scannell and Gifford 2010]. It has been argued that a “sense of place” fulfills a universal human need [Relph 1976] and promotes safety and security [Fried 2000]. In Isolated, Confined, and/or Extreme environments, place attachment can refer to the current place (e.g., spacecraft) that we may seek to support; or to the “home” environment (e.g., Earth) that we might seek to manage.	Altman, I., & Low, S. M. (Eds.). (1992). Place Attachment. Springer US. Scannell, L., & Gifford, R. (2010). Defining place attachment: A tripartite organizing framework. <i>Journal of Environmental Psychology</i>, 30(1), 1–10. Relph, Edward. Place and Placelessness (1976): In P. Hubbard, R. Kitchin, & G. Valentine, <i>Key Texts in Human Geography</i> (pp. 43–52). SAGE Publications Ltd. Fried, M. (2000). CONTINUITIES AND DISCONTINUITIES OF PLACE. <i>Journal of Environmental Psychology</i>, 20(3), 193–205.
Positive Mood	We distinguish positive mood here as a low-arousal emotion that may compose feelings of peace, tranquility, satisfaction, and calm [Brown 2021]. Closely related constructs – happiness, joy, and gratitude – are more high-arousal emotions. We choose positive mood here as the ‘catch-all’ for these positive emotions connected to the environment, as a more tangible attribute to promote as a behavioral health outcome. To distinguish from positive mood, the state of happiness is a feeling of pleasure related to the immediate environment or circumstances [Brown 2021]. Happiness is more immediate, while other factors (e.g., joy and contentment) may be more chronic, and acute happiness may not necessarily contribute to long-term contentment [Brown 2021].	Brown, B. (2021). <i>Atlas of the heart: Mapping meaningful connection and the language of human experience</i>. Random House.

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Privacy	Privacy is generally defined as the ability to regulate others' access to personal information [Westin 1968][Altman 1975]. Privacy can vary in scope: personal privacy, as a state of being private [Bailey 1979]; environmental privacy, as a characteristic of the architecture and built environment [Webster 1979]; and the person-environment relationship of privacy as a transactional and regulated process [Altman 1975]. Choice and the ability to regulate privacy is crucial. Components of privacy include the ability to be alone, as well as visual and auditory privacy [Marshall 1972].	Westin, A. F. (1968). Privacy and Freedom. Altman, I. (1975). The Environment and Social Behavior. Marshall, N. J. (1972). Human Ecology.
Psychological Resources	Resources available for psychological and emotional support. NASA provides operational psychology services for the International Space Station crew and their families. This includes private medical conferences with mental health specialists every two weeks, arranging care packages, regular cognitive function tests, and private family conferences [Sipes and Ark 2005].	Sipes, W. E., & Ark, S. T. V. (2005). Operational Behavioral Health and Performance Resources for International Space Station Crews and Families. Aviation, Space, and Environmental Medicine, 76(6).
Radiation	Radiation is the energy transmitted via waves or particles traveling through space or a medium. The annual occupational limit is 50 mSv/year; NASA's career limit is 600 mSv (corresponding to 2-3 years in Low Earth Orbit, shorter for Lunar/Martian missions), which corresponds to a 3% risk of exposure-induced death in a 35-year-old female; most other space agencies have a career limit of 1,000 mSv [NASEM 2021]. In deep space, radiation levels are elevated due to a combination of background Galactic Cosmic Radiation (GCR), Solar Particle Events (SPEs), and localized trapped particle belts. Radiation protection for spacesuits and habitats can alter the radiation dose received by the crew [Allen 2018]. GCRs are continuous and uniform flow of charged particles, arriving from outside the solar system. While GCRs have no acute effects due to their low dosage, chronic exposure is not well-characterized for long-duration spaceflight. SPEs are randomized ejections and accelerations of solar particles. The pattern of SPEs is not well understood and therefore difficult to predict. Two main electron belts and one proton belt exist around the Earth, and radiation belt models exist (e.g., AE-9, AP-9, CRRESELE, CRRESPRO). Events on Earth, such as nuclear facility accidents	Allen, C. S. et al. (2018). Spaceflight environment. In Space Safety and Human Performance. Elsevier. Jones, C. F. et al. (2013). Narrating Fukushima: Scales of a nuclear meltdown. East Asian Science, Technology and Society, 7(4). Ten Hoeve, J. E., & Jacobson, M. Z. (2012). Worldwide health effects of the Fukushima Daiichi nuclear accident. Energy & Environmental Science. National Academies of Sciences, Engineering, and Medicine. (2021). Space radiation and astronaut health: Managing and communicating cancer risks.

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	(e.g., Fukushima Daiichi, 2011), can have impacts on the safety of nearby water and food, and increased mortalities and morbidities [Jones 2013]. On a much larger scale, the use of atomic bombs (only time in history – Hiroshima and Nagasaki, 1945) resulted in over 200,000 deaths before year's end [Ten Hoeve and Jacobson 2012].	
Reconfigurability	The extent to which the habitat can be altered (physically, psychologically, or programmatically) to serve a different purpose. In a study of architectural themes across analogs and space habitats (from Apollo to the International Space Station), “flexibility” (i.e., allowing adjustments according to user requirements) was defined as a key desire, specifically [Häuplik-Meusburger 2011]: 1. Spatial flexibility (variation in size and locations) 2. Object flexibility (variation in usage) 3. Individual flexibility (personalized variations)	Häuplik-Meusburger, S. (2011). Architecture for astronauts: an activity-based approach. Springer Science & Business Media.
Resilience	Resilience has been defined as the ability to effectively negotiate, adapt to, or manage stress and trauma [Windle 2011]. The definition is two-parted: either the ability to remain psychologically healthy when faced with significant stress; or the ability to recover from a traumatic event [de Terte 2014]. A similar concept, hardiness, was introduced by Kobasa in 1982 and composed of the “belief in personal control over events, commitment to full involvement in life, enjoyment of challenge and opportunity” [Matthews 2003].	Windle, G. (2011). What is resilience? A review and concept analysis. <i>Reviews in Clinical Gerontology</i>, 21(2), 152–169. de Terte I, Stephens C, Huddleston L. (2014) The development of a three part model of psychological resilience. <i>Stress Health</i> (5):416-24. Kobasa, S. C. (1982) Commitment and coping in stress resistance among lawyers. <i>Journal of Personality and Social Psychology</i>, 42, 168–77. Matthews, G., Deary, I. J., & Whiteman, M. C. (2003). <i>Personality Traits</i>, SECOND EDITION. Cambridge University Press.
Resource Constrained	The degree of limitation of resources or resupply, including type/amount of resources and time to arrival. Resource constrained areas result from inequitable and inadequate distribution of resources (e.g., food, water, healthcare, education, mobility) [Mouratidis 2020][Bauer and Gallagher 2020]. In harsh and remote environments such as Antarctica, resources are inherently limited as they have to be transported inland, which can be logistically challenging [Kennicutt 2016]. In certain situations, medical evaluation might even be impossible [Nicholas 2016].	Mouratidis, K. (2020). Neighborhood characteristics, neighborhood satisfaction, and well-being: The links with neighborhood deprivation. <i>Land Use Policy</i>, 99. Bauer, C., & Gallagher, M. J. (2020). Education for Humanity: higher education for refugees in resource-constrained environments through innovative technology. <i>Journal of Refugee Studies</i>, 33(2), 416-436. Nicholas, M., Suedfeld, P., Weiss, K., & Gaudino, M. (2016). Affective, Social, and Cognitive Outcomes During a 1-Year Wintering in

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Factors	Definitions	Citations
Situation Awareness	Situation awareness is the internalized mental model of the current state of one's environments [Endsley 2021]. Formally, it is defined most widely as “the perception of the elements in the environment within a volume of time and space, the comprehension of their meaning and the projection of their status in the near future” [Endsley 1988]. There are three components: 1) perceiving relevant information in the environments, 2) understanding what they mean for one's goals, and 3) predicting what will happen in the system in the near future – defined as levels 1, 2, and 3 situation awareness, respectively [Endsley 2021].	Concordia. Environment and Behavior, 48(8), 1073–1091. Endsley, M. R. (1988). Design and evaluation for situation awareness enhancement. In Proceedings of the Human Factors Society 32nd Annual Meeting (pp. 97–101). Santa Monica, CA: Human Factors Society. Endsley, M. R. (2021). Situation awareness. Handbook of human factors and ergonomics, 434-455.
Sleep	Sleep is a state of reduced physical and cognitive activity, in which consciousness is altered. Adults are recommended to sleep at least 7 hours per night for optimal health [Watson 2015]. Qualifiers for healthy sleep comprise duration, quality, regularity, and timing – often defined broadly as ‘sleep architecture’ in space studies. Two separate studies on the International Space Station found that astronauts slept less than 6 hours on 35-45% of nights [Jones 2022][Barger 2014].	Watson, N. F., Badr, M. S., Belenky, G., et al. (2015). Recommended Amount of Sleep for a Healthy Adult: A Joint Consensus Statement of the American Academy of Sleep Medicine and Sleep Research Society. Journal of Clinical Sleep Medicine, 11(06), 591–592. Jones, C. W., Basner, M., Mollicone, D. J., Mott, C. M., & Dinges, D. F. (2022). Sleep deficiency in spaceflight is associated with degraded neurobehavioral functions and elevated stress in astronauts on six-month missions aboard the International Space Station. Sleep, 45(3), zsac006. Barger, L. K., Flynn-Evans, E. E., Kubey, A., Walsh, L., Ronda, J. M., Wang, W., Wright, K. P., & Czeisler, C. A. (2014). Prevalence of sleep deficiency and use of hypnotic drugs in astronauts before, during, and after spaceflight: An observational study. The Lancet Neurology, 13(9), 904–912.
Social Composition	The overall composition of the group, and includes the relevant interpersonal aspects of living and working together.	
Social Monotony	Social monotony describes the chronic exposure to the same set of people.	Das, S.K., Barhwal, K., Hota, S.K. (2015). Disrupting monotony during social isolation stress prevents early development of anxiety and depression like traits in male rats. BMC Neurosci 16, 2.

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	In general, monotony is described by an unchanging environment, repetition of events or situations, and chronic absence of novelty [Das 2015].	
Social Support	The perception or experience that one is cared for by others, esteemed and valued, and part of a mutually supportive social network [Wills 2000][Taylor 2011]. Social support can be measured via structural social support, or understanding how many interpersonal connections an individual is involved in [Wills 2000].	Wills, T. A., & Shinar, O. (2000). Measuring perceived and received social support. In S. Cohen, L. G. Underwood, & B. H. Gottlieb (Eds.), <i>Social support measurement and intervention: A guide for health and social scientists</i> (pp. 86–135). Oxford University Press. Taylor, S. E. (2011). Social support: A review. <i>The Oxford handbook of health psychology</i>, 1, 189-214.
Stress	Stress refers to any environmental, social, or internal demand which prompts an individual to adjust their usual behavior [Holmes and Rahe 1967][Thoits 1995]. We feel stressed when the demand is beyond our capability to successfully cope [Brown 2021]. Stress also has biological responses, promoting various physical and psychological effects on the body (e.g., neuroendocrine/hormone system, cerebral cortex), as well as adverse cognitive effects (e.g., executive function and task performance) [Cibrian-Llenderal 2018].	Holmes, T. H., & Rahe, R. H. (1967). The social readjustment rating scale. <i>Journal of Psychosomatic Research</i>, 11(2), 213–218. Thoits, P. A. (1995). Stress, Coping, and Social Support Processes: Where Are We? What Next? <i>Journal of Health and Social Behavior</i>, 35, 53. Cibrian-Llenderal, T., Melgarejo-Gutierrez, M., & Hernandez-Baltazar, D. (2018). Stress and cognition: Psychological basis and support resources. <i>Health and academic achievement</i>, 11(10.5772). Brown, B. (2021). <i>Atlas of the heart: Mapping meaningful connection and the language of human experience</i>. Random House.
Stress Regulation	Stress regulation is the process of responding, managing, and resolving stress. When faced with a stressor (a situation in which an individual needs to adjust their usual behavior), individuals usually vary greatly in how they regulate stress. The variables of the stress regulation process includes the magnitude of physiological/emotional responses, length of time for recovery, and the extent of adequate recovery during/following the stressor – also denoted as exposure, reactivity, recovery, and restoration [Williams 2009].	Williams, P. G., Suchy, Y., & Rau, H. K. (2009). Individual differences in executive functioning: Implications for stress regulation. <i>Annals of Behavioral Medicine</i>, 37(2), 126-140.
Supplies	Supplies encompass temporary, consumable, and depletable assets.	

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	Supplies may include personal effects, logistical support (e.g., vacuum, trash bags), scientific equipment (e.g., rodents, plants, test tubes), ECLSS (e.g., water filters, CO2 regeneration substrates), and tools. Certain supplies have been defined separately due to their relative significance, for instance food/water and medical supplies. Note that supplies are defined separately from fixtures, which is an arbitrary distinction at times. In general, we would describe supplies as items that are meant to be replaced on a more regular basis than fixtures, which may be intended to serve the same function for the lifespan of the habitat.	
Team Processes	Processes involving multiple people working together to achieve a goal beyond the capabilities of each individual alone. Proposed by Marks et al. (2001), a team moving toward a task has process “phases” when one set of actions dominates at any given time.	Marks, M. A., John E. Mathieu, & Stephen J. Zaccaro. (2001). A Temporally Based Framework and Taxonomy of Team Processes. <i>The Academy of Management Review</i>, 26(3), 356–376.
Training and Preparation	Activities (e.g., simulations, expeditions, lectures) practicing relevant skills and preparation for planned activities. The goal of training is to provide knowledge and skills to execute mission tasks, including living in the mission environment. Knowledge areas for spaceflight include introductory topics (e.g., autonomy involvement, global space sector), spaceflight systems, scientific fundamentals (e.g., aerospace engineering, life sciences, Earth sciences), operations (e.g., space health, systems and infrastructure), and special skills (e.g., robotic arm operations, extravehicular activity, fitness training, media skills, team dynamics) [Steimle and Norberg 2013]. For professional career astronauts, training can take anywhere from 3.5 to 5 years, comprising basic, advanced, and increment specific training [Steimle and Norberg 2013].	Steimle, H., Norberg, C. (2013). Astronaut selection and training. In: Norberg, C. (eds) <i>Human Spaceflight and Exploration</i>. Springer Praxis Books. Springer, Berlin, Heidelberg.
Transition Processes	Transition processes occur when teams are primarily focused on planning and evaluation of team goals. Processes most salient during the transition phase are as follows [Marks 2001]: - Mission analysis: identifying and evaluating the mission goals, environmental conditions, and resources available to execute the	Marks, M. A., John E. Mathieu, & Stephen J. Zaccaro. (2001). A Temporally Based Framework and Taxonomy of Team Processes. <i>The Academy of Management Review</i>, 26(3), 356–376.

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	mission. This includes ensuring a shared mental model of the team's purpose and objectives. - Goal specification: identification and prioritization of goals, discussing how much needs to be accomplished within a certain time frame, to what quality. - Strategy formulation: developing courses of action, including prioritization, task assignment, and discussion of expectations. Further delineating strategy formulation, this process can be broken down into 1) planning for nominal/planned activities, 2) emergency/contingency planning, and 3) reactive strategy adjustment.	
Trust	Trust is a cognitive assessment about the level of risk of making something valued, vulnerable to someone's (or something's) actions [Feltman 2011]. Trust also has been theorized to have an affective component, which is particularly important and relevant for teams (even human-robotic teams) [Jones 1996][Webber 2008][Pinney 2022]. Some aspects of trust (-building or -maintaining) include respecting boundaries, reliability, accountability, integrity, and non-judgement [Brown 2021]. These may be referred to as the interpersonal or ability-based aspects of trust, important on a personal or team level [Mayor 1995][Hoff and Bashir 2015]. Trust in automation, which is especially relevant in spaceflight contexts, are more heavily reliant on performance-based trust (i.e., performance, process, or purpose) [Lee and Moray 1992].	Feltman, C. (2011). The thin book of trust: An essential primer for building trust at work. Thin Book Publishing. Mayer R. C., Davis J. H., Schoorman F. D. (1995). An integrative model of organizational trust. <i>Academy of Management Review</i>, 20, 709–734. Hoff, K. A., & Bashir, M. (2015). Trust in automation: Integrating empirical evidence on factors that influence trust. <i>Human factors</i>, 57(3), 407-434. Lee J. D., Moray N. (1992). Trust, control strategies and allocation of function in human machine systems. <i>Ergonomics</i>, 22, 671–691. Jones, K. (1996). Trust as an affective attitude. <i>Ethics</i>, 107(1), 4-25. Webber, S. S. (2008). Development of cognitive and affective trust in teams: A longitudinal study. <i>Small group research</i>, 39(6), 746-769. Pinney, J., Carroll, F., & Newbury, P. (2022). Human-robot interaction: the impact of robotic aesthetics on anticipated human trust. <i>PeerJ Computer Science</i>, 8, e837.
Unplanned Events	Emergency, contingency, or otherwise unplanned events that usually require immediate response. The distinction between demands and events vary by context; for instance, in an hospital emergency department, what may be considered 'emergency events' are likely part of the everyday demand.	